论文与代码验证

3.1. Method

Consider two language models, A and B, and some setting in which B must perform a task where it would benefit from knowledge given to A as a prompt/encoded in A's weights (example settings in Section 4.1/Section 4.2 respectively). We propose incorporating information from A's post-layer k activation $\boldsymbol{h}_{A,k}$ into B's post-layer j activation $\boldsymbol{h}_{B,j}$ (and vice versa, though for simplicity we henceforth only discuss the first direction) (Figure 1, left).

More formally, suppose A and B (which have model dimensions d_A and d_B respectively) are given prompts x_A and x_B respectively, where x_A is of length t_A tokens and x_B is of length t_B tokens. We first run a partial forward pass of B until layer j (henceforth denoted $B_{\leq j}(x_B)$) to get $\boldsymbol{h}_{B,j} \in \mathbb{R}^{t_B \times d_B}$. Then we (1) run a partial forward pass of A until layer k to get $A_{\leq k}(x_1) := \boldsymbol{h}_{A,k} \in \mathbb{R}^{t_A \times d_A}$; (2) replace the activation of the last token $(\boldsymbol{h}_{B,j})_{t_B} \in \mathbb{R}^{d_B} \longleftarrow f((\boldsymbol{h}_{A,k})_{t_A}, (\boldsymbol{h}_{B,j})_{t_B})$ for some function $f : \mathbb{R}^{d_A+d_B} \to \mathbb{R}^{d_B}$; then (3) continue B's forward pass till decoding is complete, resulting in an output $y = B_{>k}(\boldsymbol{h}_{B,j})$.

Let $\boldsymbol{a} = (\boldsymbol{h}_{A,k})_{t_A}$, $\boldsymbol{b} = (\boldsymbol{h}_{B,j})_{t_B}$. For sake of simplicity assume $d_A = d_B$.[1] We consider three non-learned functions f:

$$f(\boldsymbol{a}, \boldsymbol{b}) = \boldsymbol{a} + \boldsymbol{b} \qquad (\texttt{sum})$$

$$f(\boldsymbol{a}, \boldsymbol{b}) = \frac{1}{2}(\boldsymbol{a} + \boldsymbol{b}) \qquad (\texttt{mean})$$

$$f(\boldsymbol{a}, \boldsymbol{b}) = \boldsymbol{a} \qquad (\texttt{replace})$$

```python
# model_ac.py
# hidden_A.shape      = [batch_size, seq_len_A, hidden_dim_A]
# hidden_states.shape = [batch_size, seq_len_B, hidden_dim_B]
def apply_hidden_A(hidden_states, hidden_A, f):
    # graft the last token activation from A
    a = hidden_A[:, -1, :]
    b = hidden_states[:, -1, :]

    d = min(a.shape[-1], b.shape[-1])
    d_A = a.shape[-1]
    d_B = b.shape[-1]

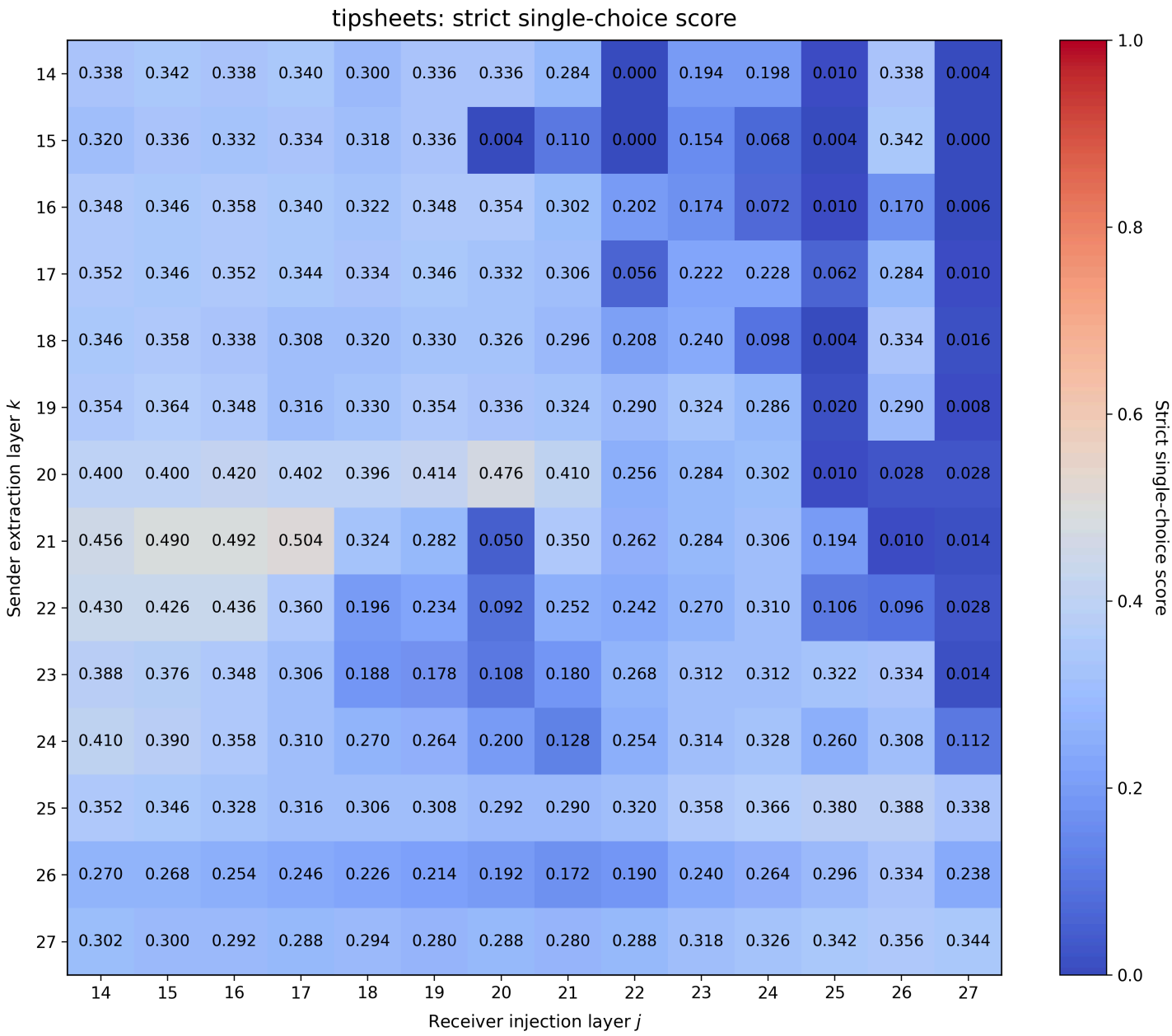
    # dimension alignment
    if f == "replace":
        new_vec = torch.cat([
            b[:, :max(0, d_B - d)],
            a[:, max(d_A - d, 0):d_A],
        ], dim=-1)
    elif f == "sum":
        new_vec = torch.cat([
            b[:, :max(0, d_B - d)],
            a[:, max(d_A - d, 0):d_A] + b[:, max(d_B - d, 0):d_B],
        ], dim=-1)
    elif f == "mean":
        new_vec = torch.cat([
            b[:, :max(0, d_B - d)],
            0.5 * (a[:, max(d_A - d, 0):d_A] + b[:, max(d_B - d, 0):d_B]),
        ], dim=-1)
    else:
        raise ValueError(f)
    hidden_states[:, -1, :] = new_vec
    return hidden_states
```

AC Qwen 7B实验结果

tipsheets任务下表现：

skyline：0.966

silent：0.312



最好的一个点(21,17)

shuffled： 0.338

zero： 0.270

countries数据集结果：

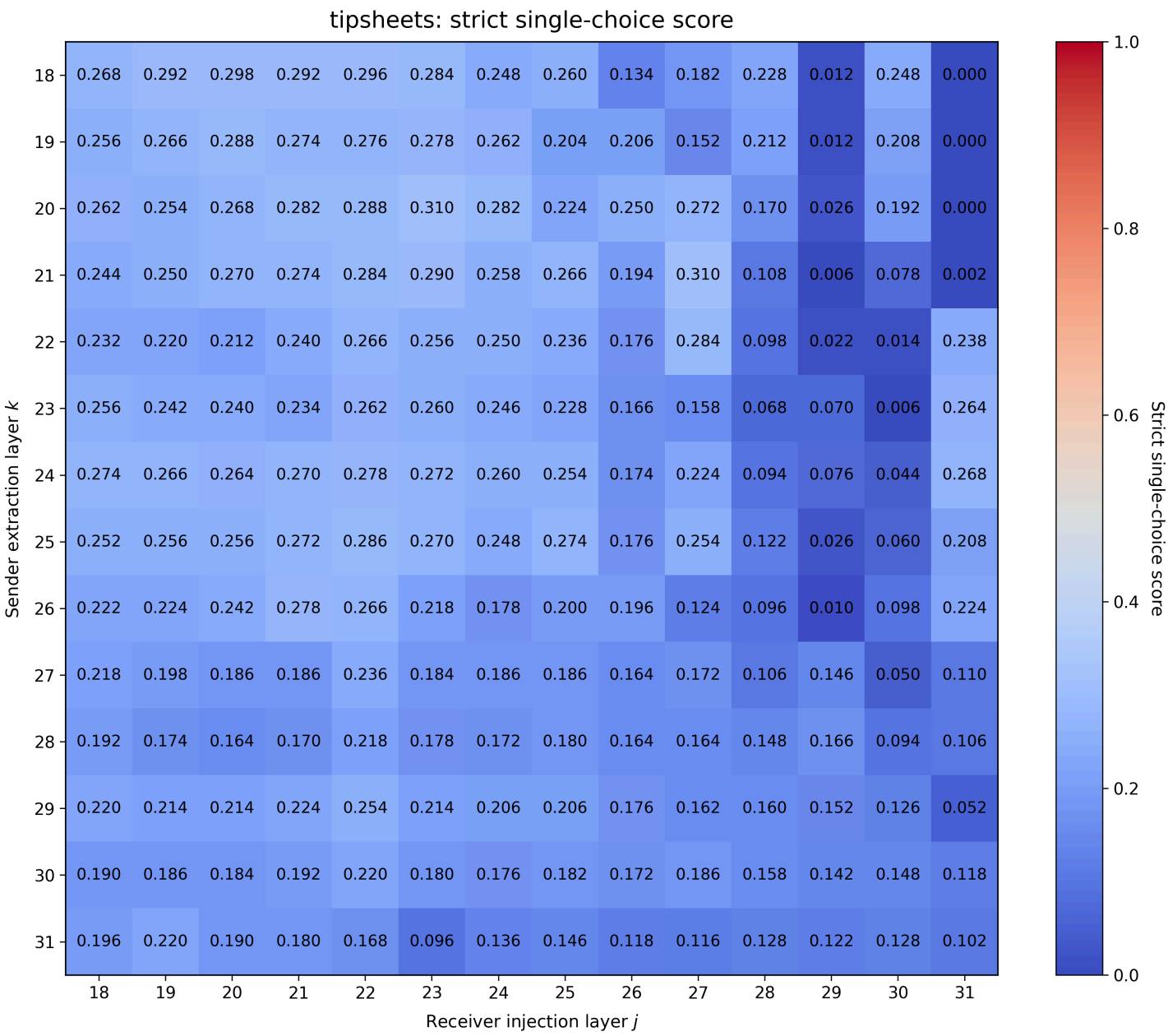
skyline：0.540

silent：0.000

(21,17):0.000

原论文模型Llama3.1 8B实验结果

skyline：0.924


tipsheets: strict single-choice score
Sender extraction layer k
Receiver injection layer j
Strict single-choice score
18 0.268 0.292 0.298 0.292 0.296 0.284 0.248 0.260 0.134 0.182 0.228 0.012 0.248 0.000
19 0.256 0.266 0.288 0.274 0.276 0.278 0.262 0.204 0.206 0.152 0.212 0.012 0.208 0.000
20 0.262 0.254 0.268 0.282 0.288 0.310 0.282 0.224 0.250 0.272 0.170 0.026 0.192 0.000
21 0.244 0.250 0.270 0.274 0.284 0.290 0.258 0.266 0.194 0.310 0.108 0.006 0.078 0.002
22 0.232 0.220 0.212 0.240 0.266 0.256 0.250 0.236 0.176 0.284 0.098 0.022 0.014 0.238
23 0.256 0.242 0.240 0.234 0.262 0.260 0.246 0.228 0.166 0.158 0.068 0.070 0.006 0.264
24 0.274 0.266 0.264 0.270 0.278 0.272 0.260 0.254 0.174 0.224 0.094 0.076 0.044 0.268
25 0.252 0.256 0.256 0.272 0.286 0.270 0.248 0.274 0.176 0.254 0.122 0.026 0.060 0.208
26 0.222 0.224 0.242 0.278 0.266 0.218 0.178 0.200 0.196 0.124 0.096 0.010 0.098 0.224
27 0.218 0.198 0.186 0.186 0.236 0.184 0.186 0.186 0.164 0.172 0.106 0.146 0.050 0.110
28 0.192 0.174 0.164 0.170 0.218 0.178 0.172 0.180 0.164 0.164 0.148 0.166 0.094 0.106
29 0.220 0.214 0.214 0.224 0.254 0.214 0.206 0.206 0.176 0.162 0.160 0.152 0.126 0.052
30 0.190 0.186 0.184 0.192 0.220 0.180 0.176 0.182 0.172 0.186 0.158 0.142 0.148 0.118
31 0.196 0.220 0.190 0.180 0.168 0.096 0.136 0.146 0.118 0.116 0.128 0.122 0.128 0.102
18 19 20 21 22 23 24 25 26 27 28 29 30 31
1.0 0.8 0.6 0.4 0.2 0.0

引用论文的实验结果

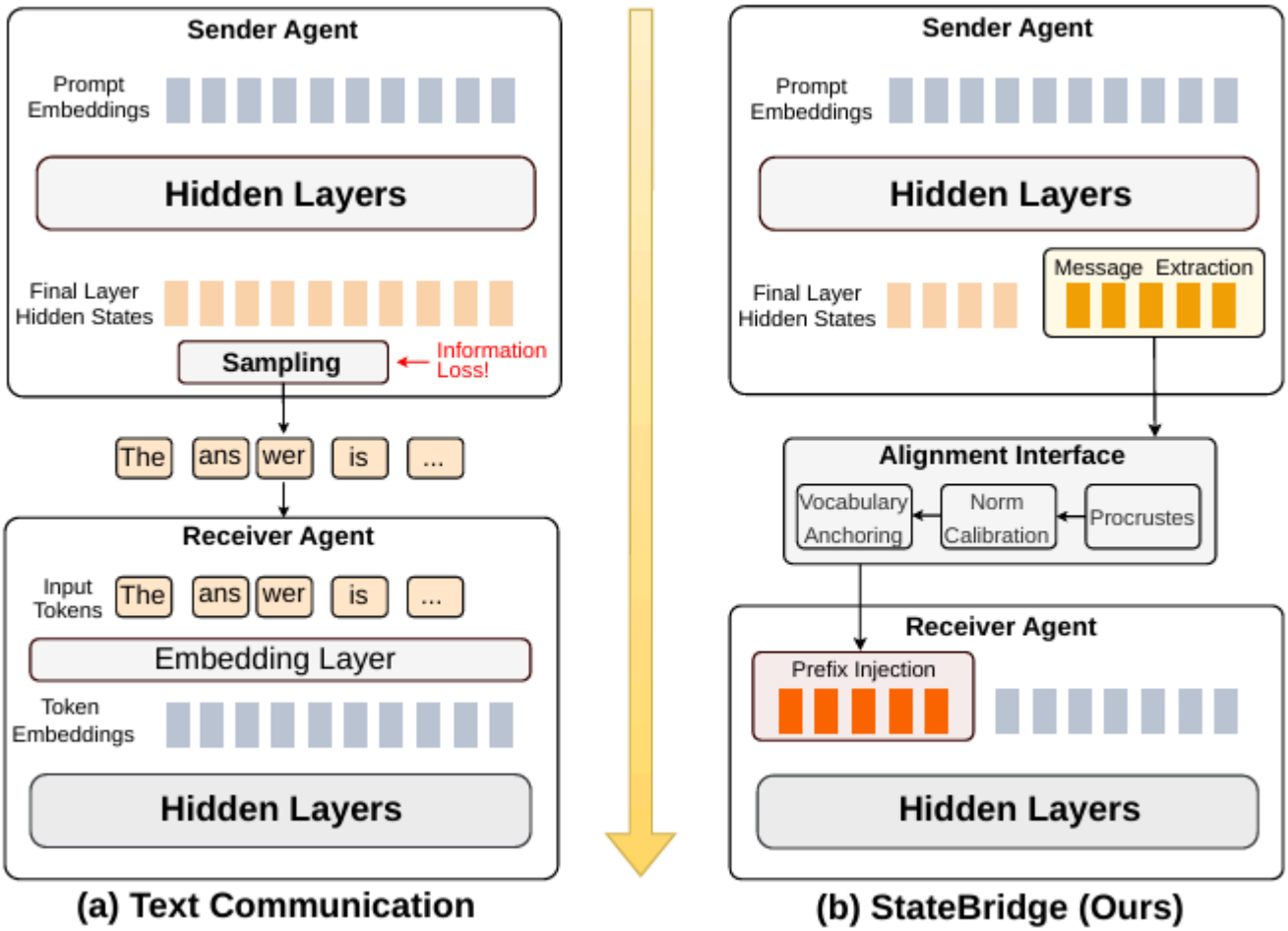
Method	Countries	Tipsheets	HotpotQA	QASPER	MuSiQuest	MultiField-QA-en	2WikiM-QA	TMATH
$\mathcal{M}_s$: huihui-ai/Llama-3.2-3B-Instruct-abliterated; $\mathcal{M}_r$: suayptalha/DeepSeek-R1-Distill-Llama-3B								
Baseline	0.05	0.32	0.23	0.05	0.02	0.11	0.27	0.34
Skyline	0.57	0.91	0.73	0.25	0.51	0.47	0.40	0.36
NLD	0.43	0.72	0.43	0.10	0.18	0.09	0.30	0.33
CIPHER	0.42	0.69	0.50	0.10	0.18	0.13	0.32	0.32
AC (mean)	0.03	0.45	0.25	0.05	0.02	0.13	0.23	**0.35**
AC (replace)	0.00	0.49	0.05	0.01	0.01	0.12	0.03	0.34
AC (sum)	0.02	0.46	0.23	0.05	0.01	0.13	0.24	0.34
KVComm (0.3)	0.46	0.45	0.46	0.09	0.28	0.15	0.28	**0.35**
KVComm (0.5)	**0.57**	**0.81**	0.57	0.27	0.32	**0.51**	0.36	**0.35**
KVComm (0.7)	**0.57**	**0.81**	**0.65**	**0.29**	**0.36**	0.47	**0.37**	**0.35**
$\mathcal{M}_s$: Orion-zhen/Qwen2.5-7B-Instruct-Uncensored; $\mathcal{M}_r$: bespokelabs/Bespoke-Stratos-7B								
Baseline	0.01	0.36	0.13	0.05	0.03	0.08	0.09	0.35
Skyline	0.51	0.97	0.53	0.10	0.25	0.40	0.09	0.35
NLD	0.21	0.80	0.16	0.02	0.04	0.11	0.02	**0.35**
CIPHER	0.04	0.60	0.03	0.01	0.03	0.07	0.03	0.34
AC (mean)	0.00	0.00	0.03	0.00	0.00	0.08	0.01	0.01
AC (replace)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
AC (sum)	0.00	0.00	0.02	0.00	0.00	0.07	0.04	0.03
KVComm (0.3)	0.04	0.26	0.02	0.01	0.01	0.09	0.08	0.31
KVComm (0.5)	0.19	0.88	0.28	0.07	0.12	0.26	0.10	0.33
KVComm (0.7)	**0.41**	**0.89**	**0.41**	**0.21**	**0.25**	**0.29**	**0.15**	0.34
$\mathcal{M}_s$: ehristoforu/falcon3-ultraset; $\mathcal{M}_r$: huihui-ai/Falcon3-7B-Instruct-abliterated								
Baseline	0.08	0.36	0.21	0.06	0.04	0.09	0.23	0.31
Skyline	0.56	0.95	0.76	0.32	0.56	0.51	0.45	0.37
NLD	**0.46**	0.80	0.52	0.19	0.25	0.11	0.24	0.15
CIPHER	0.30	0.19	0.27	0.02	0.07	0.06	0.25	0.17
AC (mean)	0.01	0.46	0.25	0.06	0.04	0.09	0.23	0.31
AC (replace)	0.00	0.49	0.12	0.00	0.01	0.13	0.17	0.31
AC (sum)	0.01	0.46	0.25	0.06	0.03	0.10	0.24	0.31
KVComm (0.3)	**0.46**	0.69	0.59	0.19	0.40	0.35	0.29	0.32
KVComm (0.5)	0.40	0.92	**0.63**	0.25	**0.44**	0.45	**0.34**	0.35
KVComm (0.7)	0.19	**0.96**	0.55	**0.26**	0.42	**0.51**	0.31	**0.36**

Statebridge论文实验结果

Published as a conference paper at COLM 2026

StateBridge: Training-free Hidden-state Alignment for Latent Communication in LLM Multi-Agent Systems

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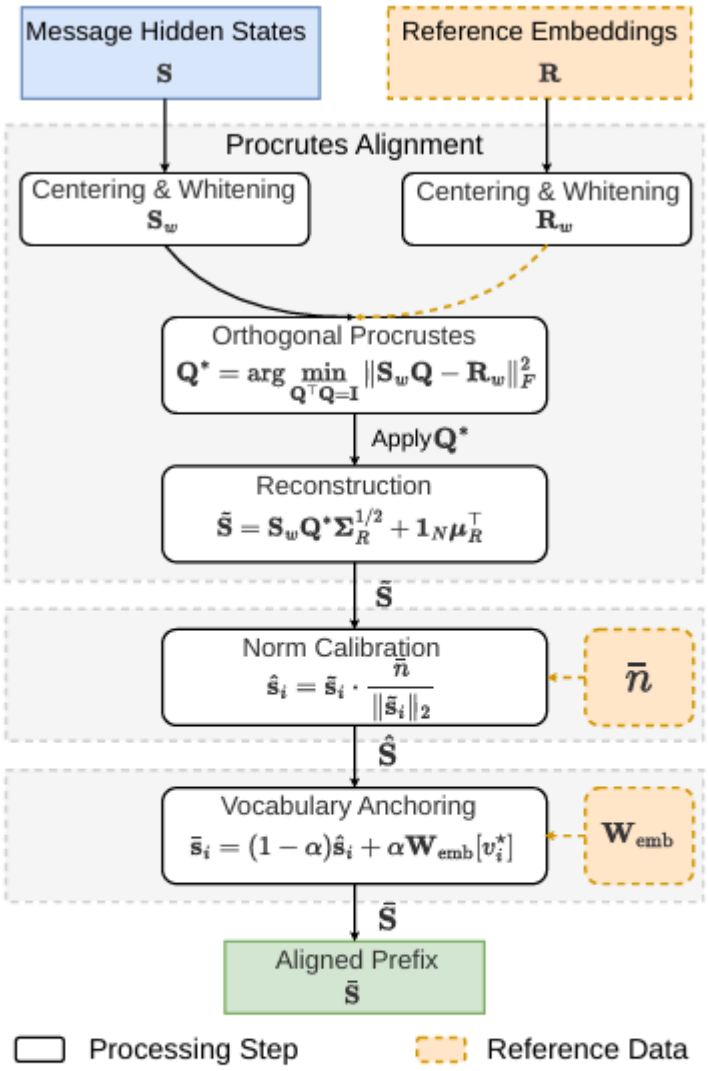


Figure 2: The StateBridge alignment interface transforms message hidden states $\mathbf{S}$ into the aligned prefix $\bar{\mathbf{S}}$ for receiver using reference embeddings $\mathbf{R}$.

prefix token=64


```
"completed_pairs": 100,
  "modes": {
    "none": {
      "original_accuracy": 0.28,
      "swapped_accuracy": 0.32,
      "both_correct_rate": 0.0,
      "invalid_count": 0
    },
    "raw_text": {
      "original_accuracy": 0.96,
      "swapped_accuracy": 0.98,
      "both_correct_rate": 0.95,
      "invalid_count": 0
    },
    "tail_text": {
      "original_accuracy": 0.83,
      "swapped_accuracy": 0.88,
      "both_correct_rate": 0.81,
      "invalid_count": 1
    },
    "tail_embeddings": {
      "original_accuracy": 0.88,
      "swapped_accuracy": 0.86,
      "both_correct_rate": 0.84,
      "invalid_count": 0
    },
    "statebridge": {
      "original_accuracy": 0.88,
      "swapped_accuracy": 0.86,
      "both_correct_rate": 0.84,
      "invalid_count": 0
    },
    "mismatched": {
      "original_accuracy": 0.2,
      "swapped_accuracy": 0.18,
      "both_correct_rate": 0.0,
      "invalid_count": 94
    }
    }
```

prefix token=16


```
"none": {
    "original_accuracy": 0.32,
    "swapped_accuracy": 0.32,
    "both_correct_rate": 0.0,
    "invalid_count": 0
  },
  "raw_text": {
    "original_accuracy": 0.98,
    "swapped_accuracy": 1.0,
    "both_correct_rate": 0.98,
    "invalid_count": 0
  },
  "tail_text": {
    "original_accuracy": 0.3,
    "swapped_accuracy": 0.5,
    "both_correct_rate": 0.12,
    "invalid_count": 0
  },
  "tail_embeddings": {
    "original_accuracy": 0.36,
    "swapped_accuracy": 0.44,
    "both_correct_rate": 0.14,
    "invalid_count": 0
  },
  "statebridge": {
    "original_accuracy": 0.36,
    "swapped_accuracy": 0.46,
    "both_correct_rate": 0.16,
    "invalid_count": 0
  },
  "mismatched": {
    "original_accuracy": 0.36,
    "swapped_accuracy": 0.3,
    "both_correct_rate": 0.0,
    "invalid_count": 6
  }
```


实验发现tail_embeddings和statebridge的结果几乎一样，进一步检查发现，答案分布也是类似的，statebridge即使经过全部的transfomer处理和上下文的融合，但未能有效利用上下文信息？hidden states经过对齐变换反而退化成了token embedding表示

向量	与对应词嵌入的平均余弦相似度	最近词为原 token 的比例
原始隐藏状态	0.0370	0%
对齐后，关闭词汇锚定	**0.9891**	**100%**
完整 StateBridge	**0.9944**	**100%**

余弦相似度越接近 1，方向越接近。这里的"最近词"是从**整个模型词表**中搜索，不是只在输入 token 中搜索。

shape-like/representation alignment

AC,Statebridge

function-like/prefix adapter

Interlat,ThoughtComm